THE UNIVERSITY OF HONG KONG
FACULTY OF BUSINESS AND ECONOMICS

AI for Business Decisions

GENERAL INFORMATION

Instructor: Dr. Shan Huang
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Consultation times: By appointment

Tutor: TBA
Pre-requisites: None
Co-requisites: None
Mutually exclusive: N/A
Course Website: Moodle

Reference Book: Recommended readings listed below
Programming language: Python (via Google Colab)
Software: Google Colab and ChatGPT

COURSE DESCRIPTION

Artificial Intelligence (AI) is fundamentally reshaping how organizations make decisions. This course provides a rigorous, decision-oriented introduction to AI for business, emphasizing not only how AI models are built, but how they are evaluated, deployed, and governed in real organizational settings. The course integrates machine learning, experimentation, and generative AI into a unified decision framework. Through lectures, hands-on coding exercises, real-world business cases, and team projects, students develop the technical literacy and managerial judgment needed to assess AI applications, avoid common pitfalls, and design AI-driven solutions with measurable business impact.

COURSE OBJECTIVES

The purpose of this course is to provide a comprehensive introduction to AI and practical applications of AI in the context of business decision-making.

After taking this course, students should be able to:
1. Design AI-driven solutions to optimize business processes and support strategic decision-making across diverse industries.
2. Apply supervised, unsupervised, and generative AI techniques to analyze large-scale business data and extract actionable insights.
3. Distinguish between offline model performance and online business impact, and design appropriate evaluation strategies, including A/B testing.
4. Integrate AI technologies into existing organizational workflows while accounting for operational constraints and incentives.
5. Critically evaluate AI applications in business, including their limitations, unintended consequences, and ethical implications.
6. Assess issues of fairness, bias, interpretability, and transparency in AI-driven decisions and propose principled remedies.

PROGRAMME LEARNING OUTCOMES

PLO1: Acquisition and internalization of knowledge of the programme discipline

PLO2: Application and integration of knowledge

PLO3: Inculcating professionalism and leadership

PLO4: Developing global outlook

PLO5: Mastering communication skills

COURSE LEARNING OUTCOMES

Course Learning Outcomes	Aligned Programme Learning Outcomes
CLO1: Explain AI workflows and core supervised, unsupervised, and generative AI techniques for business decision-making. CLO2: Evaluate AI model performance and business impact using offline metrics, online experimentation, and A/B testing. CLO3: Design AI-driven solutions for real business problems in areas such as pricing, customer management, operations, advertising, and LLM-enabled decisions. CLO4: Assess governance, fairness, bias, interpretability, transparency, and ethical implications in AI-driven decisions.	PLO1, PLO2 PLO1, PLO2 PLO2, PLO4, PLO5 PLO3, PLO4

COURSE TEACHING AND LEARNING ACTIVITIES

Course Teaching and Learning Activities	Expected contact hour	Study Load (% of study)
T&L1. Interactive lectures, real-world cases, and hands-on coding exercises	39	28%
T&L2. Moodle readings, preparation, and online engagement	25	18%
T&L3. Individual assignments and quizzes	46	33%
T&L4. Team project and peer collaboration	30	21%
Total	140	100%

Assessment Methods	Brief Description (Optional)	Weight	Aligned Course Learning Outcomes
A1. Individual assignments	Five applied assignments using AI methods, evaluation, and business interpretation	50%	CLO1, 2, 3, 4
A2. Team project	Team project applying course knowledge to a real business problem; slides and short video	20%	CLO3, 4
A3. Individual quizzes	Short closed-book quizzes; lowest three quiz scores dropped	20%	CLO1, 2
A4. Class participation	Quality of in-class contributions (5%) and Moodle online discussions (5%)	10%	CLO2, 3, 4
	Total	100%	

STANDARDS FOR ASSESSMENT

Course Grade Descriptors

A+, A, A-	• Demonstrate a strong understanding of AI workflows, evaluation methods, and business applications. • Handle questions professionally and contribute original, well-supported arguments. • Excellent performance in assignments, quizzes, project deliverables, and participation, with accurate analysis and clear business implications.

B+, B, B-	• Demonstrate a good understanding of relevant AI concepts and business decision frameworks. • Handle questions logically and contribute ideas that go beyond lectures and assigned materials. • Good performance in assignments, quizzes, project deliverables, and participation, with accurate computation and sound analysis.
C+, C, C-	• Demonstrate a basic understanding of the concepts involved. • Address questions in an acceptable and reasonably structured manner. • Acceptable performance in assignments, quizzes, project deliverables, and participation, with reasonably accurate computation and analysis.
D+, D	• Demonstrate only minimum understanding of the concepts involved. • Address questions only marginally and show limited participation. • Marginally acceptable performance in assignments, quizzes, project deliverables, and participation, with noticeable gaps in computation, analysis, or business interpretation.
F	• Demonstrate poor understanding of the concepts involved. • Unable or unwilling to handle questions or participate constructively. • Fail to meet passing standards for assignments, quizzes, project deliverables, or participation, with major analytical or integrity problems.
Assessment Rubrics for Participation	
A+, A, A-	• High-quality contributions to in-class and online discussions. • Consistently prepared, respectful, and responsive to peers and instructors. • Contributions demonstrate strong understanding and connect AI concepts to business decisions.
B+, B, B-	• Good participation in class and Moodle discussions. • Often prepared and respectful in interactions. • Contributions demonstrate good understanding and logical engagement with course topics.
C+, C, C-	• Some participation in discussions. • Sometimes prepared and engaged. • Contributions show basic understanding but limited depth or consistency.
D+, D	• Minimal or irregular participation. • Rarely prepared for discussion or collaboration. • Contributions show only minimum understanding of course concepts.
F	• Minimal or no participation in class or online discussions. • Almost never prepared or engaged. • Unable or unwilling to contribute constructively or follow course expectations.
Assessment Rubrics for Assignments, Quizzes, and Team Project	
A+, A, A-	• Work is accurate, complete, and submitted on time. • Applies AI techniques and evaluation methods rigorously. • Shows originality, clear reasoning, and strong connection to business impact.
B+, B, B-	• Work is complete and generally accurate. • Applies AI techniques and evaluation methods appropriately. • Shows clear reasoning and good connection to business implications.
C+, C, C-	• Work meets basic requirements with acceptable accuracy. • Applies course concepts in a generally understandable way. • Shows some connection to business implications but limited depth.
D+, D	• Work is incomplete, late where penalties apply, or only marginally accurate. • Applies course concepts with significant gaps. • Shows weak analytical reasoning or limited connection to business implications.

F	• Work fails to meet basic requirements or is not submitted. • Shows major conceptual, computational, or analytical errors. • Fails to demonstrate understanding of AI methods or violates course integrity expectations.

COURSE CONTENT AND TENTATIVE TEACHING SCHEDULE

No	Topic	Content
AI Pipeline		
1	Introduction to AI	What is AI The AI workflow
2	Algorithms and Data	Training and learning Data engineering
3	Offline Evaluation	Offline model evaluation Metrics and applications
4	Online Evaluation I	A/B Testing fundamentals
5	Online Evaluation II	A/B Testing Statistics
6	Online Evaluation III	A/B Testing Advanced
7	Industrial Applications: A/B Testing	Guest lecture: A/B testing
AI for Business Decisions		
1	AI for Pricing	Toyota Pricing Case
2	AI for Managing Customers	Bookbinders Case
3	Explainable AI	Loan Approval Case
4	AI for Managing Operations	Store Traffic Case
5	AI for Advertising	Guest lecture: AI advertising
6	LLMs for Business Decisions	Agentic Commerce Generative AI Economic Impact

** Time spent may vary across topics.*

REQUIRED/RECOMMENDED READINGS & ONLINE MATERIALS (e.g. journals, textbooks, website addresses etc.)

Lecture slides, assigned readings, and cases will be posted on Moodle before or after class.
Recommended textbooks:
• Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, and Jonathan Taylor, Introduction to Statistical Learning (ISLR). ISBN-13: 978-3031387463; ISBN-10: 3031387465. Available at https://hastie.su.domains/ISLP/ISLP_website.pdf.download.html.
• Ron Kohavi, Diane Tang, and Ya Xu, Trustworthy Online Controlled Experiments: A Practical Guide to A/B Testing. Cambridge University Press, 2020. https://exp-platform.com/.
Students must also have access to Google Colab and ChatGPT throughout the course.

MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE

x conducting mid-term survey in addition to Student Feedback on Teaching and learning (SFTL) around the end of the semester
x Online response via Moodle site
O Others: _____ (please specify)

COURSE POLICY (e.g. plagiarism, academic honesty, attendance, etc.)

Students are expected to participate actively, engage respectfully with peers' ideas, and contribute meaningfully to team-based learning activities.
The use of generative AI tools (e.g., ChatGPT, Gemini) is encouraged for learning, exploration, and experimentation. Students remain fully responsible for understanding, validating, and explaining all submitted work, and all use of AI tools must comply with university academic integrity policies.
Academic dishonesty, including cheating, plagiarism, unauthorized collaboration, falsifying academic records, or misrepresenting work, is unacceptable.

ADDITIONAL COURSE INFORMATION (e.g. e-learning platforms & materials, penalty for late assignments, etc.)

Students should check Moodle at least three times per week for announcements and deadlines.
Individual assignments will be posted on Tuesdays and are due on Sundays during Weeks 2-6. Each assignment must be submitted on Moodle on time; late submissions will not be accepted and there is no grace period.
The team project will be posted on Moodle in Week 6 and will be due in two weeks. Deliverables include project slides and a short video. A peer evaluation component may add or deduct points based on teammates' evaluations.
Short, closed-book quizzes will be given at the end of each class session. Some quizzes serve as attendance or engagement checks, while others assess conceptual understanding. The lowest three quiz scores will be dropped.